

The Great Divide

Finding Things Faster with Binary Search

Lesson Summary: How do computers search through millions of items in a fraction of a second? Instead of checking item by item (Linear Search), efficient algorithms use **Binary Search**—a "divide and conquer" strategy that repeatedly cuts the search area in half.

1. Linear Search vs. Binary Search

Imagine looking for a specific item. Depending on how your data is organized, you can choose between two main search strategies:

Linear Search (One-by-One)

How it works: Check every single item starting from the beginning until you find what you need.

Requirement: Works on *any* list (sorted or unsorted).

Analogy: Searching for a spice jar in an messy market stall or flipping page-by-page from page 1.

Speed: Slow for large datasets (up to N tries for N items).

Binary Search (Divide & Conquer)

How it works: Jump to the middle, check if your target is higher or lower, and eliminate half the options!

Requirement: MUST be on a **SORTED** list (e.g., numerical or alphabetical order).

Analogy: Opening a dictionary right to the middle to locate a word.

Speed: Extremely fast (takes at most ~7 tries for 100 items!).

2. Why Must the List Be Sorted?

If a list is unsorted (e.g., a random list of student names), looking at the middle element tells you *nothing* about where the other names are located. Binary Search fails completely!

When roll numbers (1–60) or names (A–Z) are sorted, knowing that your target is greater than the middle item guarantees that you can safely **discard the entire lower half**. *"Sorting is the secret key to Binary Search's speed!"*

3. The 4-Step Binary Search Algorithm

Step 1 Find the Middle: Locate the middle item of your current sorted list range.

Step 2 Compare with Target: Is the middle item your target? If yes, you found it! 🎉 If not, determine if your target is higher or lower than the middle item.

Step 3 Discard Half: Eliminate/cross out the entire half where your target cannot possibly exist. You instantly cut your remaining work in half!

Step 4 Repeat: Repeat Steps 1–3 on the remaining half until the target item is found.

4. Real-World Applications

Algorithms like Binary Search power software and services we interact with daily:

- **IRCTC & Railway Reservation Systems:** Searching millions of PNR status records instantly using sorted database indexes.
- **Search Engines (Google):** Rapidly retrieving web pages matching search queries out of billions of indexed documents.
- **Online Dictionaries & Phonebooks:** Finding contact details or word definitions without scanning every entry.



Key Takeaways:

- Binary search **requires** sorted data.
- Each comparison cuts remaining possibilities by **50%**.
- Linear search checks 1-by-1; Binary search divides and conquers!

Student Worksheet & Activity

Interactive Exercises on Binary Search Principles

Name: _____

Activity 1: Guided Walkthrough — The Library Challenge

Scenario: Our school library has 100 books numbered consecutively from **1 to 100**, placed in order on a long shelf. We need to find **Book #72** using Binary Search.

Step #	Current Search Range	Middle Book Checked	Comparison (Higher/Lower?)	Action Taken
Guess 1	1 to 100	Book #50	Target (#72) is Higher	Discard books 1–50
Guess 2	51 to 100	Book #75	Target (#72) is Lower	Discard books 76–100
Guess 3	51 to 74	Book #62	Target (#72) is Higher	Discard books 51–62
Guess 4	63 to 74	Book #68	Target (#72) is Higher	Discard books 63–68
Guess 5	69 to 74	Book #71	Target (#72) is Higher	Discard books 69–71
Guess 6	72 to 74	Book #73	Target (#72) is Lower	Discard books 73–74
Guess 7	72 to 72	Book #72	Match Found! 🎉	Search Complete

Notice: Even in the worst-case scenario for 100 items, Binary Search took only 7 steps, whereas Linear Search could have taken 72 steps!

Activity 2: Concept Reinforcement Exercises

Question 1: Unsorted Data Challenge

Suppose you have an unsorted list of 8 animal names: [Zebra, Monkey, Elephant, Cat, Dog, Bear, Lion, Fox]. Can you perform a Binary Search to find "Dog"? Explain why or why not, and describe what step must be taken first.

Question 2: Efficiency Comparison Table

Fill in the maximum number of guesses required in the worst-case scenario for Linear Search versus Binary Search for different dataset sizes:

Total Items in List (N)	Linear Search (Max Guesses)	Binary Search (Max Guesses)
10 items	10 guesses	4 guesses
100 items	100 guesses	7 guesses
1,000 items	1,000 guesses	_____ guesses
1,000,000 items	1,000,000 guesses	20 guesses

Question 3: Real-World Scenario Identification

For each of the following scenarios, state whether **Linear Search** or **Binary Search** would be more appropriate and why:

- a) Finding a contact name in an alphabetically ordered contact list on a smartphone.
- b) Finding a lost blue sock in an unsorted pile of laundry.

Practice Questions for Students

Grade 6 • Homework / Quiz

Independent Assessment & Problem Solving

Instructions: Answer the following questions clearly. Show your work step-by-step for calculation and trace questions.

1. Multiple Choice Questions

a) What is the primary prerequisite requirement before you can run a Binary Search algorithm on a list?

- A) The list must contain only even numbers.
- B) The list must be sorted in order.
- C) The list must have fewer than 50 items.
- D) The target item must be right in the middle.

b) If you double the size of a sorted list from 100 elements to 200 elements, how many additional steps does Binary Search take in the worst case?

- A) 100 extra steps
- B) 50 extra steps
- C) 1 extra step
- D) It doubles the number of steps

2. Binary Search Algorithm Trace

Consider the following sorted array of 9 numbers:

[4, 12, 19, 25, 33, 47, 58, 62, 81]

You are searching for the target value **58** using Binary Search.

Task: List each step of the search process. For each step, state:

- The current search range (sub-array)
- The middle element chosen
- Whether the target is higher or lower than the middle element
- Which half is eliminated

3. Short Answer & Conceptual Thinking

a) Why is Binary Search classified as a "Divide and Conquer" algorithm? What specific action represents "dividing" and what represents "conquering"?

b) Imagine you are playing the Number Guessing Game (1 to 100). The secret number is **17**. Write down the sequence of guesses a player using Binary Search would make until they guess 17 correctly.

4. Real-World Application Challenge

An e-commerce website has 10,000 products stored in its database. When a customer searches for Product ID #4,521, explain why the server uses Binary Search instead of Linear Search, and calculate the maximum number of checks the server will perform before finding the product.